

PART A – Usability in Design (usability.gov ideas)

1. Big idea: 3 things a good site must satisfy

Your slides show a simple triangle:

1. **Meeting the organization's objectives**
2. **Meeting usability goals** for the site
3. **Meeting users' needs**
4. **Complying with basic web guidelines**

The process is:

- First, **review** the site: is it meeting business goals + usability goals?
- Have a **kickoff meeting** and set **measurable usability goals** for your site or app.

👉 **MEMORIZE:** These phrases:

- “Meeting organization's objectives & usability goals”
- “Meeting users' needs”
- “Complying with basic web guidelines”

These are the 3 angles your teacher can ask.

2. Measurable usability goals

2.1 What is a measurable usability goal?

A measurable usability goal = definition of **successful usability** for a **specific set of users** doing a **specific task**.

Not vague like “Site should be easy.” It’s concrete and testable.

2.2 Example: Mobile tax app (from slide)

Scenario:

You build a **mobile app** for **small business owners** to pay **state withholding taxes** online. Previously they mailed a check; now they’ll pay online. On the **first attempt**, you might set goals like:

The business owner will:

1. **Successfully complete the transaction**
 - In **≤ 5 minutes**
2. **Submit the right amount** from the **right bank + account**
3. **Make no more than one error**
 - And recover from any error in **≤ 1 minute**
4. Rate the experience **4 or 5** on a **1–5 scale**
5. Indicate they **want to use the app again**.

This is a perfect exam example of a **measurable usability goal**.

👉 **MEMORIZE:** “Specific users + specific task + measurable success criteria (time, accuracy, errors, satisfaction, future use).”

2.3 Four main kinds of usability measures

Your slides list 4 big categories: **time, accuracy, overall success, satisfaction**.

We’ll break them down.

2.3.1 Time

We can set goals for:

- Overall **time to complete** the task
- Or break it into:
 - **Time to get to the application / right info section**
 - **Time to use** the application / understand information
 - **Time to recover from an error**

Example:

“90% of users will be able to find the tax-payment page in less than 2 minutes and complete payment in less than 5 minutes.”

2.3.2 Accuracy

We can set goals for:

- **Number of unproductive navigation choices** (wrong clicks, wrong paths)
- **Number of unproductive searches** (searches that don't give what they need)
- **Number of errors using an application** (wrong data entered, wrong button clicked)
- **Number of misunderstandings of information** (e.g., misreading a policy)

Example:

“Users will make no more than 1 unproductive navigation choice during checkout.”

2.3.3 Overall success

Core idea: **Users must actually succeed.**

If users cannot do their tasks or cannot get answers to their questions on your app, then your app is failing for those users for those tasks/questions.

We can also make success-related goals, like:

- New users:
 - Will go to **Help** if needed
 - Will **find what they need** in Help
 - Will **return to their original task within 2 minutes**
 - Returning users:
 - Who have done the task before should **do it successfully again without Help**
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2.3.4 Satisfaction

Users must be **happy** – not just successful.

We can measure:

- Overall satisfaction
- Or break it down:
 - Satisfaction with **navigation**
 - Satisfaction with **search**
 - Satisfaction with **level of detail** in content
 - Satisfaction with **language** (clarity, tone)
 - And other specific factors

👉 **MEMORIZE: 4 usability metrics = Time, Accuracy, Success, Satisfaction.**

2.4 Relying on measures (performance vs satisfaction)

Important slide.

When testing your site against usability goals:

- Treat **performance (time, accuracy, success)** as **more important** than satisfaction.
- Why?
 - Users often give **high satisfaction ratings** even when they had serious problems.
 - They may:
 - **Blame themselves** (“I’m bad with computers”)
 - Not want to hurt your feelings
 - Be polite

So:

- **Low ratings** → definitely fix the site.
- **High ratings** → may be misleading, check performance data too.

👉 **MEMORIZE:** Performance > satisfaction when they conflict.

2.5 Examples of measuring usability – levels

Your slides divide goals into:

- **Website level**
- **Scenario level**
- **Page level**

Website level examples

- 95% of customers will be able to **find and order a product**.
- 95% of physicians will be able to **find, read, and understand** latest lung cancer treatment info.

- 95% of travelers will be able to **make their own airline reservations**.
- All trained service reps will handle **25 customer calls per hour**.

Scenario level examples

Scenario = multi-page, a full interaction.

- 90% of users will **find a specific article** on thyroid cancer within 3 minutes.
- 90% of users will **read an update article** in less than 5 minutes.
- 90% will **make an airline reservation** in less than 5 minutes.

Page level examples (often homepage)

- 90% of users will be able to **find and click a specified link** within 15 seconds.
- 90% will **find and click a specified graphic** within 2 seconds.
- Page will **download in ≤ 5 seconds** on broadband.

👉 **MEMORIZE:** These 3 levels + 1 example each (website, scenario, page).

3. Meeting user's needs

Your site should be checked against **what users actually want/need**, not just business goals.

Ways to review:

- Review **types of users** and **reasons they come** to the site.
- Review user **emails and phone calls** (feedback).
- Evaluate **web logs and search logs** (what are they searching? where do they drop?).
- Conduct an **online survey**.
- Conduct a **usability test**.

So you align the design with:

- What users search for
 - What they complain about
 - What they actually do on the site (logs)
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Tips for usability test (from this deck)

- **Test the site, NOT the users**
 - Users are not “stupid”; if they fail, it’s usually a design issue.
- **Performance vs subjective measures** – we already saw: performance is more reliable.
- **Make use of what you learn**
 - Actually fix the design, don’t just collect data and ignore.
- **Find the best solution**
 - Iterate: test → improve → test.

👉 **MEMORIZE:** “Testing the site, not the users” – classic exam line.

4. Complying with basic web guidelines

Three tasks:

1. Conduct a **heuristic evaluation**
2. Review **research-based usability & web design guidelines**
3. Examine **organization’s internal policies**

So you mix:

- Established external guidelines (Nielsen, W3C, etc.)
 - Internal brand/policy rules
 - Your own heuristic evaluation findings
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4.1 Heuristic Evaluation – definition

A usability inspection method where a **small set (1–3) of evaluators** examine a user interface and judge its compliance with a set of **usability principles (heuristics)**.

Key points:

- 1–3 evaluators (more than 1 but not a crowd).
- They work **independently**, then results are combined.
- Output = **list of potential usability problems**.

👉 **MEMORIZE** this definition + “1–3 evaluators independently checking against heuristics.”

4.2 Nielsen’s 10 heuristics (from slide)

You MUST know these by name:

1. **Visibility of system status**
2. **Match between system and the real world**
3. **User control and freedom**
4. **Consistency and standards**
5. **Error prevention**
6. **Recognition rather than recall**

7. **Flexibility and efficiency of use**
8. **Aesthetic and minimalist design**
9. **Help users recognize, diagnose, and recover from errors**
10. **Help and documentation**

👉 **MEMORIZE ALL 10 NAMES.** This is high-probability exam/quiz content.

4.3 Gerhardt-Powals heuristics

These are more cognitive/engineering style heuristics. Two slide chunks:

- **Automate unwanted workload**
- **Reduce uncertainty** – display data clearly and obviously
- **Fuse data** – summarize lower-level data into higher-level (e.g., show “average speed” instead of 100 raw readings)
- Present new info with **meaningful aids to interpretation**:
 - Use **familiar frameworks** (everyday terms, metaphors)
 - Use names **conceptually related** to function, context-dependent
 - **Improve recall and recognition**
- Group data in **consistently meaningful ways** to decrease search time
- **Limit data-driven tasks** – reduce time spent reading raw data
- Make appropriate use of **color and graphics**
- Include **only info needed at the moment**
- Provide **multiple coding** of data when appropriate
- Practice **judicious redundancy** (repeat key info in different ways, but not spammy)

👉 For exam: **MEMORIZE 3–4 of these** and be able to explain them in plain words.

4.4 Web Accessibility (WAI)

- **Web Accessibility Initiative (WAI)** is a W3C initiative: [w3.org/WAI](https://www.w3.org/WAI).
- Focus: guidelines so people with disabilities can access the web (WCAG etc.)

👉 **MEMORIZE:** WAI = W3C initiative for web accessibility.

PART B – Usability Testing Extended (Evaluation from controlled to natural settings)

Now the second file is more about **methods and environments**:

- Usability testing
 - Experiments vs usability testing
 - Remote testing
 - In-the-wild studies
 - Long-term studies
 - Participant numbers, etc.
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5. Usability testing – what happens concretely

Usability testing:

- Tests **how people perform on products in controlled settings**.
- People are observed while doing tasks.
- We record:
 - Time to complete tasks
 - Number and types of errors.
 - Video, keystrokes, mouse movements.

Data is used to:

- Calculate performance times
 - Identify and explain **errors**
 - Measure **user satisfaction** via questionnaires/interviews
 - Sometimes include observations in **more natural contexts**, even in-the-wild.
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5.1 Quantitative measures used in usability testing

Slides list typical measures:

- Number of participants who **successfully complete** the task
- Time to complete the task
- Time to complete the task **after time away** from it (learning/retention)
- Number and type of errors per task
- Number of errors per unit of time
- Number of times people navigate to an item like **online help**
- Number of people making the same/similar errors

👉 **MEMORIZE:** at least 4–5 quantitative measures.

6. Usability labs and equipment

Slides show:

- **Usability lab** where designers watch users through a **one-way mirror** and record sessions.
- Layout example: US Health & Human Services usability lab.
- Eye-tracking devices like **Tobii Glasses** for mobile eye-tracking.

Equipment trend:

- Getting **smaller and more portable**.
 - New models of eye-tracking glasses look like regular glasses.
 - Video/audio often recorded using **mobile phones**.
 - This replaces bulky devices and tripods and is more discreet.
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7. Remote testing

Remote testing = participants carry out tasks in **their own environment** without evaluator physically present.

Tools:

- Zoom, Teams, etc.
- Testing can be:
 - **Synchronous** (live session)

- **Asynchronous** (participants complete tasks on their own; logs + recordings are reviewed later)

Advantages:

- Multiple participants tested **at the same time**, each in their own environment.
- Easier for **people with disabilities** (no travel).
- Covid-19 pushed creative remote setups.

Downsides:

- No one physically there to help with **equipment problems**.
- Less **personal** than in-person testing.

👉 **MEMORIZE:** At least one advantage and one disadvantage.

8. Case 1 – iPad usability (Budiu & Nielsen, 2010)

This is a **classic usability study**.

Context:

- Before iPad launch, study was done in two cities: Fremont, CA and Chicago, IL.
- Needed:
 - Quick results for third-party app devs
 - To be done **secretly** so competitors wouldn't know.
- Participants: 7 people with **over 3 months** iPhone experience.

Procedure:

- Participants signed **informed consent**, stating:

- What tasks they'd do
- Length of study
- Compensation
- Right to **withdraw any time**
- Promise of **confidentiality** and no identity disclosure
- They were asked to **explore the iPad**, then perform specified tasks (randomly assigned).

Examples of tasks: (slide shows a list like reading news, using apps, etc.).

Problems found:

- Difficulties accessing the web
- Lack of affordances & feedback (not clear what is tappable)
- Getting lost in an application
- Not knowing where to tap

Actions:

- Problems reported to developers
- Findings made publicly available by Nielsen Norman Group.
- Later follow-up study: studied how iPad used in everyday life (because first study didn't cover that).

👉 **MEMORIZE:** At least 3 problem types (getting lost, unclear tappareability, lack of feedback).

9. Case 2 – Remote testing for XR (extended VR)

Context:

- Study at **Krone Corp (Finland)** for new VR/XR collaborative platforms.
- Covid-19 → needed remote testing across **multiple countries** (Finland, India, China, Germany, Indonesia, Malaysia, USA, UAE).

Setup:

- Needed sophisticated remote testing for experts and employees.
- Some tests done on participants' premises with **special disinfecting routines**.
- Others fully remote.
- Very little prior advice on how to do such remote XR testing during a pandemic.

The slides then show:

- Photos of participants with headsets
- Screenshots of VR environments
- Testing setup diagrams

This mainly illustrates **complex remote usability setups**.

10. Experiments vs usability tests (quick recap)

Slides repeat some content from Week 8 but summarize:

Experiments:

- Test hypotheses
- Predict relationship between 2+ variables
- IV manipulated by researcher
- DV measured

- 1 or 2 IVs typically
- Validated statistically
- Replicable

Experimental designs:

- **Different participants** (between participants) – each group in different condition
- **Same participants** (within participants) – everyone experiences both conditions
- **Matched participants** (pairwise) – participants matched by characteristics (like expertise, gender) and one of each pair assigned to each condition.

Slides also show a diagram of different vs same vs matched (nothing new conceptually).

11. In-the-wild studies

Definition:

- Done in **natural settings**, where prototypes are used **freely**.
- Less controlled than lab or traditional field study.
- Goal: understand what users do naturally and how technology affects them.

Used in product design to:

- Identify new tech opportunities
 - Determine design requirements
 - Decide how to introduce new technology
 - Evaluate tech in real use situations
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Case: Painpad (pain monitoring device)

Context:

- Monitoring patients' pain after surgery is challenging.
- Painpad: a tangible keypad for self-logging pain.
- First usability-tested in lab, then brought to **two UK hospitals** for in-the-wild study.

Participants & data:

- Two studies; 54 people (13 males, 41 females); mean age around 64.
- Privacy important.
- Patients got Painpad after surgery and were **prompted every 2 hours** to log pain.
- Nurses also collected scores and logged them.
- Data gathered:
 - Satisfaction questionnaires (Likert 1–5)
 - Compliance with 2-hour routine
 - Comparison: Painpad vs nurse-collected scores

Findings (from slide):

- Satisfaction high (average ~4.63/5).
- Compliance mixed: some liked using it, others ignored or disliked prompts.
- Patients recorded **more pain scores** with Painpad than through nurses.

👉 Shows how in-the-wild studies give **realistic usage patterns** that lab can't show.

12. Long-term studies

Some evaluations leave designs with participants for **extended periods** (2–4 weeks):

- Especially for **complex technical products** like visualization tools.
- People need time to integrate them into their **own work**.

Data collection:

- Initial interview
 - Daily-use diaries
 - Automatic logging
 - Follow-up interviews, etc.
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13. How many participants are enough?

For usability testing:

Depends on:

- Schedule
- Participant availability
- Cost

Typically:

- **5–10 participants** if possible.
- Some experts: test until **no new insights** (saturation).
- Others: **5 participants** is enough to uncover most serious problems.

For other evaluations (experiments, in-the-wild):

- Experiments: number depends on type of experiment; often need many → consult statistician.
- In-the-wild: could be a few people, a team, or a whole community.

👉 **MEMORIZE:** “5–10 for usability tests, 5 often enough to find serious issues.”

14. Usability testing vs experiments (comparison table)

Usability Testing:

- Goal: **Improve products**
- Few participants
- Results **inform design**
- Not always completely replicable
- Conditions controlled **as much as possible**
- Procedure planned, results reported to **developers**

Experiments (for research):

- Goal: **Discover knowledge**
- Often **many participants**
- Results **validated statistically**, must be **replicable**
- Strongly controlled conditions
- Experimental design, scientific report to **scientific community**

👉 **MEMORIZE this contrast.** Very exam-able.

15. Summary slides (just reinforcing)

- Usability testing = controlled spaces (labs, temporary labs), focus on performance and errors.
 - Collect indirect observation (video, logs), plus questionnaires and interviews.
 - Remote testing uses portable equipment and became important in Covid.
 - Long-term studies evaluate complex designs over weeks.
 - Experiments manipulate IVs and measure DVs.
 - In-the-wild studies deploy prototypes in real settings; findings can be unexpected.
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PRACTICE: Week 10 Questions

A. 5 MCQs (with answers + explanations)

MCQ 1

Which of the following is the **best definition** of a **measurable usability goal**?

- A. A general statement that the site should be easy to use.
- B. A definition of successful usability for any user doing any task.
- C. A definition of successful usability for a **specific set of users** doing a **specific task**, expressed in measurable terms.
- D. A guarantee that users will not make mistakes.

Correct answer: C

Explanation:

A measurable usability goal is **specific and testable**, tied to particular users, tasks, and measurable results like time, accuracy, or satisfaction.

MCQ 2

Which of the following is **NOT** listed as one of the four main types of **usability measures**?

- A. Time
- B. Accuracy
- C. Overall success
- D. Color scheme

Correct answer: D

Explanation:

The four main types are **time, accuracy, overall success, and satisfaction**. Color scheme affects usability but is not one of the four measurement categories in this model.

MCQ 3

In a **heuristic evaluation**, which statement is TRUE?

- A. A large group of more than 20 users evaluates the interface.
- B. Evaluators follow a rigid script of tasks.
- C. A small set of evaluators independently judge compliance with usability principles.
- D. Only quantitative measures of time and errors are collected.

Correct answer: C

Explanation:

Heuristic evaluation is a **usability inspection method** where typically 1–3 evaluators independently inspect the UI against heuristics and report problems.

MCQ 4

Which of the following is an **advantage of remote usability testing** mentioned in the slides?

- A. It always provides better video quality.
- B. Multiple participants can be tested at the same time in their own environments.
- C. There are no equipment problems.
- D. It completely replaces in-person testing.

Correct answer: B

Explanation:

Remote testing makes it possible to test several participants **simultaneously** in their own settings. It does not eliminate equipment problems and does not fully replace in-person tests.

MCQ 5

According to your slides, which statement correctly contrasts **usability testing** and **experiments**?

- A. Usability testing aims to discover new scientific knowledge, while experiments aim to improve products.
- B. Usability testing usually involves many participants, while experiments involve very few.
- C. Usability testing focuses on improving products with few participants, while experiments aim to discover knowledge and require statistically validated, replicable results.
- D. Both are identical; the only difference is the type of lab used.

Correct answer: C

Explanation:

Usability testing is usually conducted with **few participants**, aims to **improve products**, and results inform design. Experiments aim to **discover knowledge**, often use more participants, and must be statistically validated and replicable.

B. 5 Short Answer Questions (with model answers)

SAQ 1

List and briefly explain the four main categories of measurable usability goals.

Model answer:

The four categories are:

1. **Time** – how long users take to perform tasks, including time to find the right page, complete the task, and recover from errors.
2. **Accuracy** – how correctly users perform tasks, including number of errors, unproductive navigation, and misunderstandings.
3. **Overall success** – whether users can actually complete tasks or get answers to their questions, possibly with help.
4. **Satisfaction** – whether users are happy with the experience, including navigation, search, content detail, and language.

SAQ 2

Why do your slides say that performance measures are more important than satisfaction ratings in usability testing?

Model answer:

Performance measures (time, accuracy, success) are more reliable because users often give **high satisfaction ratings even when they had serious problems**. They may blame themselves, be polite, or not want to hurt the evaluator's feelings. Low ratings definitely indicate a problem, but high ratings can be misleading, so designers should prioritize performance data.

SAQ 3

What is the main purpose of an in-the-wild study, and how is it different from lab-based usability testing?

Model answer:

In-the-wild studies place prototypes in **natural settings** so users can use them freely as part of their real lives. The goal is to understand what users do naturally and how technology affects them. This is different from lab-based usability testing, which is done in **controlled environments** on predefined tasks, focusing on performance metrics like time and errors.

SAQ 4

Describe what happens in a heuristic evaluation and name two sets of heuristics your slides mention.

Model answer:

In a heuristic evaluation, a **small group of evaluators (usually 1–3)** independently examines a user interface and judges how well it complies with a set of usability principles (heuristics). They produce a list of potential usability problems. The slides mention **Nielsen's heuristics** (e.g., visibility of system status, consistency and standards) and **Gerhardt-Powals heuristics** (e.g., automate unwanted workload, reduce uncertainty, fuse data).

SAQ 5

According to your slides, how many participants are typically used in usability testing, and why might this number be considered enough?

Model answer:

Typically, **5–10 participants** are used in usability testing, and some experts say that **5 participants** are enough to identify most serious usability problems. This is because many critical issues are common across users, and after a certain number of participants, fewer new problems are discovered, so further testing gives diminishing returns.

C. Case Study (with full answer)

Case Study

You are responsible for redesigning a **university course registration website**.

The university wants to:

1. Set **measurable usability goals** for the new site.
2. Test the site with students in a **lab-based usability test**.
3. Later, observe how the site is used in **real life**, after launch.

Question:

1. Propose **two measurable usability goals**: one at **scenario level** and one at **page level**, using the time/accuracy/success/satisfaction framework.
 2. Describe how you would conduct a **lab-based usability test** (mention at least three quantitative measures to collect).
 3. Describe one way you could later perform an **in-the-wild study** of the registration site.
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Model Answer

1. Measurable usability goals

- **Scenario-level goal** (multi-page task):

“90% of students will be able to **search for and register in all their required courses** in less than **10 minutes**, making **no more than one navigation error**, and successfully completing registration with no missing courses.”

Here we define success (registering required courses), time (≤ 10 minutes), and accuracy (≤ 1 navigation error).

- **Page-level goal** (within a single page, e.g. homepage or course search page):

“90% of students will be able to **find and click the ‘Register for Courses’ link** on the homepage within **15 seconds**.”

This is similar to the “find a specified link in 15 seconds” example in the slides.

These goals use **time, accuracy, and success**.

2. Conducting a lab-based usability test

To test the new site:

- **Participants:** Recruit representative students (e.g., undergrads from different years).
- **Environment:** Use a **usability lab** or a quiet controlled room with screen recording.
- **Tasks:** Ask them to carry out realistic tasks like:
 - “Register in courses A, B, and C.”
 - “Drop a course and add a different one.”

Quantitative measures to collect (at least three):

1. **Time to complete each task** (e.g., time to finish registration).
2. **Number and type of errors per task** (e.g., adding wrong section, going to wrong page, failed submissions).

3. **Success/failure rate:** Number of participants who successfully complete the task.
4. (Optional extra) **Number of times they navigate to Help/FAQ** and whether Help helps them recover.

Additionally, collect **satisfaction ratings** via a short questionnaire (e.g., 1–5 scale for ease of use and satisfaction).

3. In-the-wild study after launch

After the site is live, you can perform an **in-the-wild study** by:

- Deploying the site to all students as the main registration tool.
- Observing usage via:
 - **Web logs** (where users go, how often they backtrack, where they drop out).
 - Optional **on-site surveys** asking students about their real-life experiences.
- You might also:
 - Interview a sample of students during registration week.
 - Watch how they use the system in a **computer lab** or on their own laptops during real registration, while they are under real stress and time pressure.

This gives you a picture of how the system works in **natural conditions**, beyond controlled lab testing.